

THE NORTH CAROLINA SCHOOL OF SCIENCE AND MATHEMATICS
NATIONAL SOCIETY OF BLACK ENGINEERS



SMATHFORCE

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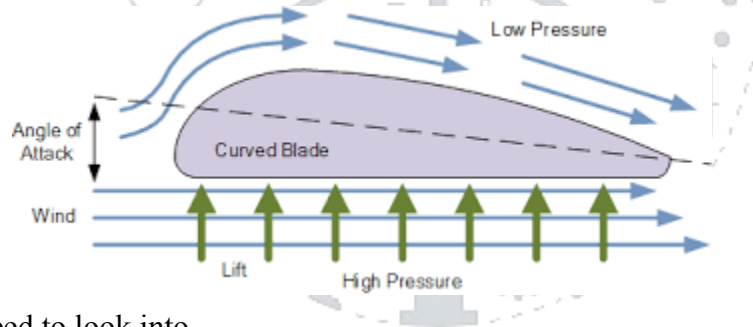
BLADE DESIGN

Material should be:

- Relatively Lightweight (Heavier provides more torque but requires more wind)
- Stiff

Blades should be:

- 5 or 7 blades
- Long as possible
- Tapered towards the tips
- Possibly include “tubercles.”
- 20-degree pitch
- Most likely flat blades
- Blades should be able to be disassembled(screwed in dowels)
- 3d-printed dowels
- 3d printed glue in



Still need to look into

- Width
- Twist
- ONSHAPE LINK:

<https://cad.onshape.com/documents/22463c911f2fb193153bc7d2/w/c0d18d145525028712dc574a/e/327745d8314173daea751c07?renderMode=0&uiState=67886482c86d9422f1cb4b80>

GEAR DESIGN

SHAFT

- 1.563IN INSIDE DIAMETER OF THE PVC PIPE
- 0.164IN FOR THICKNESS OF THE DESIGN

Material:

- Aluminum (relatively lightweight for strength)
- Plastic (Easy to 3D print)
- Steel? (Very sturdy but also heavy which reduces efficiency)

Type of Gears:

- Spur gears: **Description:** The most straightforward type of gear with straight teeth parallel to the axis of rotation.
Advantages: Easy to work with, widely available, and efficient for transferring power between parallel shafts.
- Helical Gears: **Description:** Smooth gears that generate little to no noise.
Advantages: Smoother gears = effective rotations and faster speeds
[Wind Turbines: How Gears Help Harness Renewable Energy - TechMeStuff](#)
- Gear ratio explained:
https://www.youtube.com/watch?v=txQs3x-UN34&ab_channel=driving4answers

MORE INFORMATION ON COMMONLY USED GEARS: [Wind Turbine Gears: Design and Applications](#)

<https://globecore.com/publications/wind-turbine-gearbox-types/#:~:text=2,>

<https://www.ever-power.net/spur-gears-in-wind-turbines/>

<https://www.ever-power.net/product-category/gear/spur-gear/>

Gear Ratio:

- Higher ratio needs more torque
- Higher gear ratio optimizes RPM
- ~~Ratios between 5:1 and 10:1 are typically used (7:1 is most common)~~
- For Kidwind 4:1 to 6:1 are ideal

- 3D print gears first and test
- Scrap exchange for PVC Pipes for Base
- A 7:1 ratio often works well if the generator needs ~1000 RPM.
- **Low Wind Speeds:** Lower gear ratios (e.g., 5:1) help maintain torque.
- **High Wind Speeds:** Higher gear ratios (e.g., 8:1 to 10:1) maximize RPM.
- Goal is to get to around 1000 rpm and we will use whatever gear ratio gets us there.

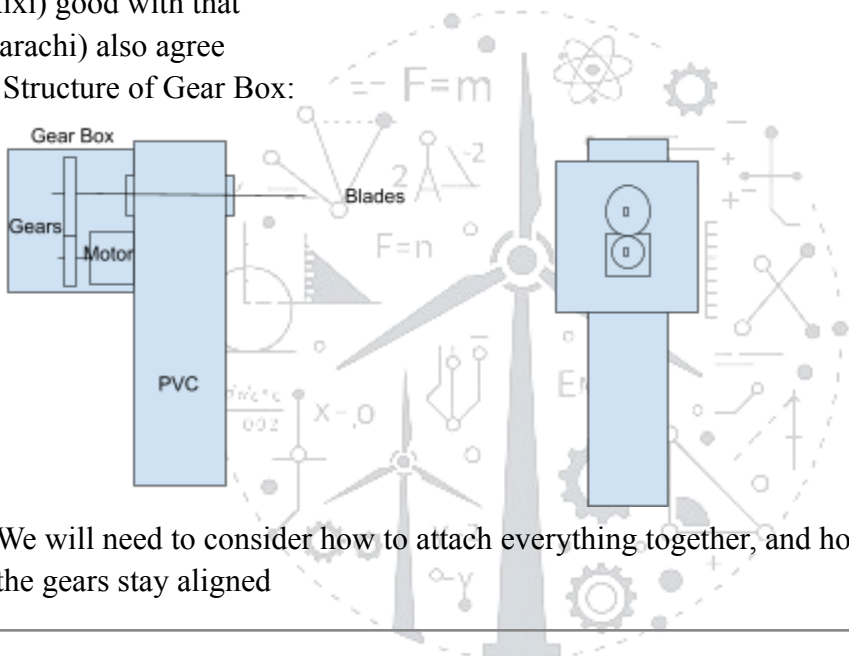
Arrangement:

- I (Paige) think that we should just use two gears rather than using multiple to build up a larger gear ratio, but if we decide we need a very large gear ratio then we should reconsider so that our gears remain a reasonable size.

^ I'm (xixi) good with that

^ I (Amarachi) also agree

Overall Structure of Gear Box:

- 
- We will need to consider how to attach everything together, and how to make sure that the gears stay aligned

Generators

Generator Type	Optimal Gear Ratio	Comments
DC Motor (12V)	6:1 to 8:1	Ideal for moderate wind speeds.
Stepper Motor (NEMA 17)	7:1 to 10:1	Works well with high-speed blade designs.
PMDC Generator (Bike Dynamo)	5:1 to 7:1	Great for consistent, medium wind speeds.

Servo Motor (Modified)

8:1 to 10:1

Best for experimental or lightweight setups.

What to Look for in a Generator**1. Voltage and Current Output:**

- Select a generator that can produce sufficient voltage (~6V–12V) at your turbine's operating speed.

2. RPM Range:

- Choose a generator optimized for high RPMs (e.g., 1000–3000 RPM) when using higher gear ratios.

3. Ease of Integration:

- Ensure the generator can be mounted and connected easily to your gear system.

4. Durability:

- Use a generator that can handle varying wind conditions without overheating or failing.

Known measurements:Kidwind Wind Speed: $v_w = 3.5 - 5 \text{ m/s}$

Our Wind speed: 2.0 m/s

Weight of the blades:

Version 1: 242g

Version 2: 229g

Black Nose/Hub:

- weight: 73g
- Slits for wings depth: 0.738 inches

Green Nose/Hub weight: 81g

Driver Gear(120 teeth):

Diameter: 4.7705 inch

Gear attached to generator:

Diameter: 0.472 inch

Generator:

Diameter of center piece: 0.7405 inch

Axle diameter: 0.0775

Length: 1.5 inch

Axles diameter:

Axles: 3/16th of an inch

PVC: 2 inches

Torque measurements:

- Hole distance from the center: 39.5cm

Bearings:

- Diameter: 0.4975 in
- Distance from hole to edges: 0.1525in
- Thickness: 0.1935 in

Hub Measurements: $4.2\text{cm} = 0.042\text{m}$

Distance to wing center of mass from Hub: $15.9\text{cm} = 0.159\text{m}$

Distance to wing center of mass from Axle center: $r = 20.1\text{cm} = 0.201\text{m}$

Blade Angular Velocity: $\omega = 10.9\text{ rad/s} = 1.735\text{ rev/s}$ @ wind speed of 2.0 m/s

Blade Efficiency: $c_d = 35\%$

- ρ : Density of air (kg/m^3)

Known equations:

- Blade power = $0.15 \times \text{Diameter}^2 \times \text{windspeed}^3$
 - Blade power = $0.15 \times \text{Diameter}^2 \times \text{windspeed}^3$
 - = $0.15 \times (2.4\text{ metres})^2 \times (10\text{ metres/second})^3$
 - = $0.15 \times 6 \times 1000 = 900\text{ watts approx.}$
 - (2.4m diameter rotor at 10 metres/sec or 22 mph)

- Wind Energy: $P = \frac{1}{2} \rho A v^3$
 - P : Power (W)
 - ρ : Density of air (kg/m^3)
 - A : Swept Area (m^2)
 - v : Wind Speed (m/s)

- $V(\text{volts}) \times A(\text{amps}) = W(\text{watts})$
- *Inverse relationship between torque and speed

120

1. **First Stage: 1:3 Reduction**
 - Driver Gear: **12 teeth**
 - Driven Gear: **36 teeth**
 - **(12:36 → 1:3)**
2. **Second Stage: 1:4 Reduction**
 - Driver Gear: **12 teeth**
 - Driven Gear: **48 teeth**
 - **(12:48 → 1:4)**
3. **Third Stage: 1:5 Reduction**
 - Driver Gear: **10 teeth**
 - Driven Gear: **50 teeth**
 - **(10:50 → 1:5)**
4. **Fourth Stage: 1:2 Reduction (Final Output to Generator)**
 - Driver Gear: **15 teeth**
 - Driven Gear: **30 teeth**
 - **(15:30 → 1:2)**

Total Reduction Calculation

- **1st stage:** 1:31:31:3
- **2nd stage:** 1:41:41:4
- **3rd stage:** 1:51:51:5
- **4th stage:** 1:21:21:2
- **Total Gear Ratio:** $3 \times 4 \times 5 \times 2 = 120$ $3 \times 4 \times 5 \times 2 = 120$

Determining where to place gearbox:

On the shaft, the driver gear is located at 21 inches up

The second axle needs to be placed 5.11 inches from the original hole with the driver gear(will put calculations for image)

$21 \text{ in} - 5.11 \text{ in} = 15.89 \text{ inches}$ on the shaft

The hole on the

1500 rpm for 3 volts

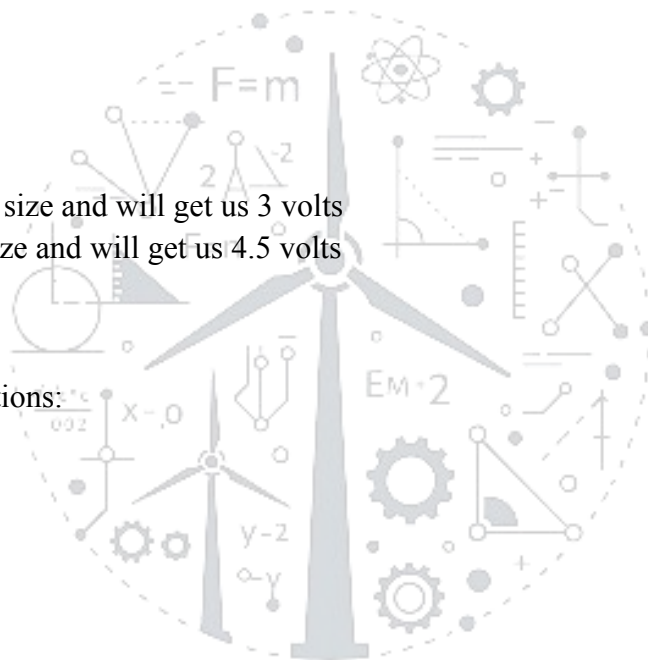
2500 rpm for 4.5 volts

102rpm

New gear ratio options

- 1;15 - good for size and will get us 3 volts
- 1;25- bad for size and will get us 4.5 volts

1:25 gear ratio calculations:



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Assuming 21 inches up
 diameter of axle: $3/16$ inches = 0.1875

gear grooves: 0.075 in

gear connected to generator: 0.472 inches

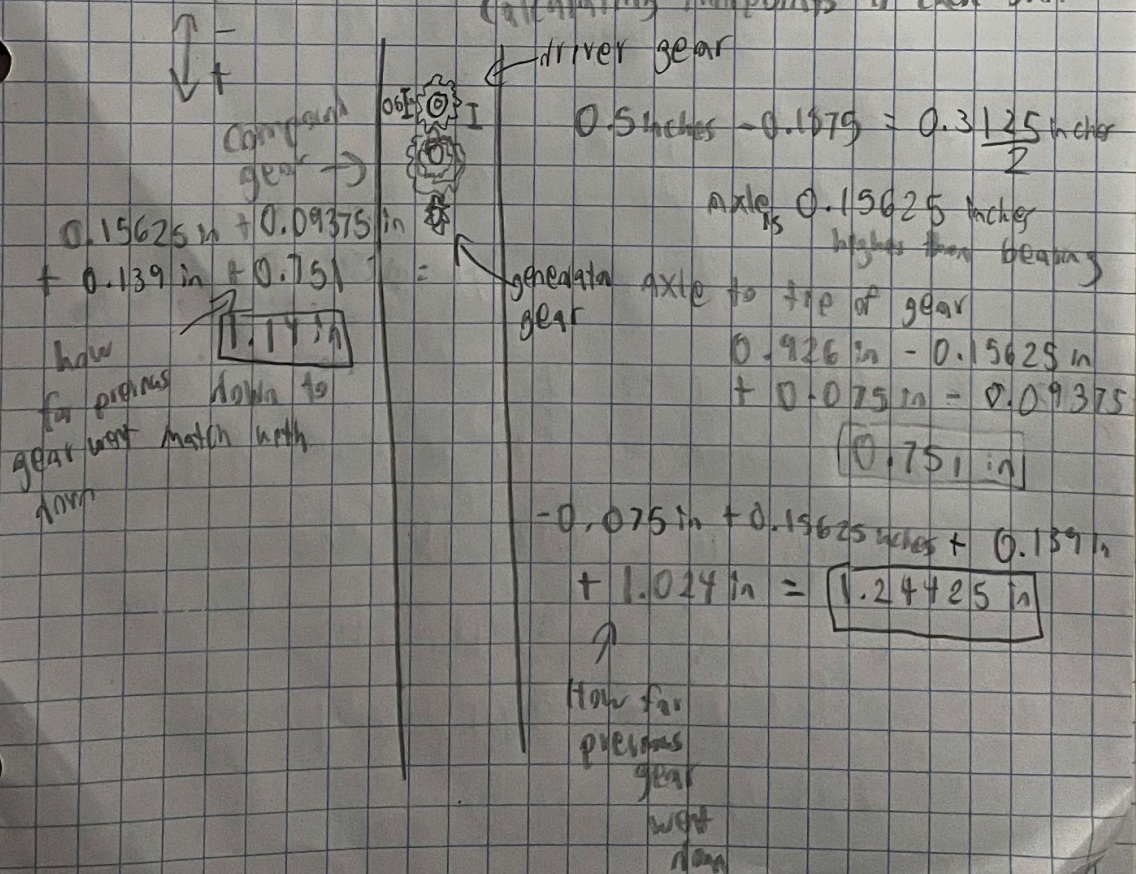
Stand alone gear: 2.047 inches diameter

Compound gear big part: 2.047 inches

Compound gear small part: 0.472 inches

bearing diameter: 0.5 inches

calculating midpoints of each shaft



Works cited:

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[Wind-turbine-blade-design](#)

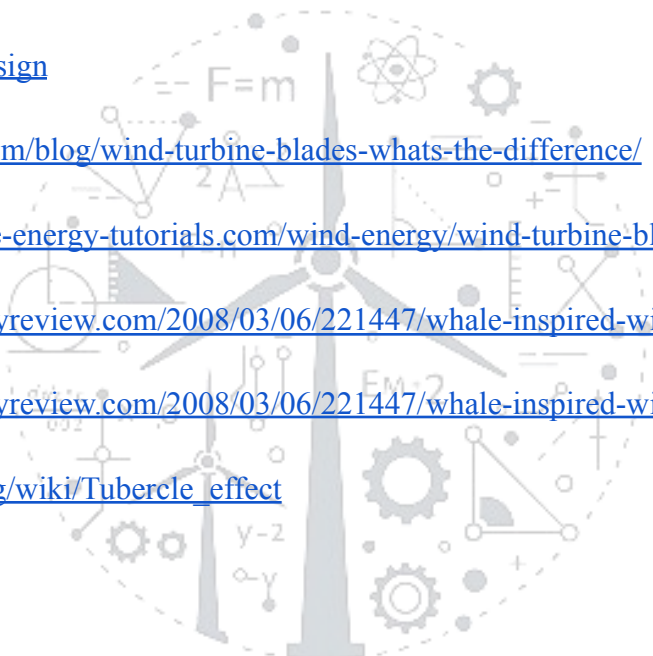
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